

Übungsseite: <https://mathe.aufgabenfuchs.de/potenz/logarithmus.shtml>

Übungsbeispiele:

Aufgabe 1: Logarithmus

Verwandle folgende Potenzgleichungen in Logarithmengleichungen:

- | | | | | |
|---------------|----------------|---------------------------|----------------------------|--------------------|
| a) $2^6 = 64$ | c) $4^4 = 256$ | e) $8^1 = 8$ | g) $10^{-3} = 0,001$ | i) $36^{0,5} = 6$ |
| b) $3^3 = 27$ | d) $9^0 = 1$ | f) $3^{-1} = \frac{1}{3}$ | h) $2^{-5} = \frac{1}{32}$ | j) $243^{0,2} = 3$ |

Aufgabe 2: Logarithmus

Verwandle folgende Logarithmengleichungen in Potenzgleichungen

- | | | | | |
|---------------------|---------------------|--------------------|---------------------|---------------------------------|
| a) $\log_3(81) = 4$ | c) $\log_6(36) = 2$ | e) $\log_8(1) = 0$ | g) $\log_3(9) = 2$ | i) $\log_6(3) = 0,5$ |
| b) $\log_4(64) = 3$ | d) $\log_2(64) = 6$ | f) $\log_5(5) = 1$ | h) $\log_7(49) = 2$ | j) $\log_{64}(2) = \frac{1}{6}$ |

Aufgabe 3: Logarithmus

Berechne die folgenden Logarithmen und mache die Probe

- | | | | | |
|-----------------|-----------------------|-----------------------|--------------------|---------------------------------------|
| a) $\log_2(16)$ | c) $\log_{10}(10000)$ | e) $\log_{10}(0,001)$ | g) $\log_2(0,5)$ | i) $\log_3\left(\frac{1}{27}\right)$ |
| b) $\log_3(27)$ | d) $\log_{10}(0,1)$ | f) $\log_5(0,2)$ | h) $\log_2(0,125)$ | j) $\log_2\left(\frac{1}{256}\right)$ |

Aufgabe 4: Logarithmengesetze

Vereinfache und berechne soweit wie möglich:

- | | | |
|---|--|--|
| a) $\log_2(a) - \log_2(b)$ | d) $(\log_3(a^2))^{-3} + (\log_3(1))^3$ | g) $\log_3(x^2) + \log_3\left(\frac{1}{x^2}\right)$ |
| b) $\log(x+y) - \log(x)$ | e) $\frac{1}{2} \log(4) + 3 \cdot \log(6) - 2 \cdot \log(3 \cdot 2^3)$ | h) $-\frac{1}{3} \log(x^2 y^{-2} z) + \frac{1}{3} \log(x^{-1} y z)$ |
| c) $\log_3\left(\frac{b}{c}\right) + \log_3(b)$ | f) $2 \cdot \log(x) + \frac{1}{2} \log(x^4) - \log(x^2)$ | i) $\log_4 \sqrt{a} + \log_4 \sqrt{(ab)^{-1}} + \frac{1}{2} \cdot \log_4(b)$ |

Aufgabe 5: Logarithmengesetze

Vereinfache und berechne soweit wie möglich mit dem Taschenrechner

- | | | | | |
|-------------------------------------|----------------------|--------------------------------------|-------------------------------------|--------------------------------------|
| a) $\log_3(5)$ | c) $\log_{1,1}(100)$ | e) $\log_2(5^3)$ | g) $\log_2\left(\frac{x}{8}\right)$ | i) $\log_3 \sqrt[3]{a}$ |
| b) $\log_2\left(\frac{1}{3}\right)$ | d) $\log_4(24)$ | f) $\log_2\left(\frac{32}{3}\right)$ | h) $\log_3(s^{-1,5})$ | j) $\log_3\left(\frac{1}{ay}\right)$ |

Aufgabe 6: Logarithmengleichungen

Bestimme Definitions- und Lösungsmenge der folgenden Gleichungen durch Exponieren

- | | | | |
|---------------------|----------------------|--|--------------------------------|
| a) $\log_3(x) = 2$ | e) $\log_5(x) = 0$ | i) $2 \cdot \log_3(x) - \log_3(x+6) = 0$ | m) $\log_2(x^2 - 1) = 3$ |
| b) $\log_4(x) = 4$ | f) $\log_4(x) = 0,5$ | j) $\log_2(x) + \log_2(x+2) = 3$ | n) $\log_3(x^2 + 2x + 2) = -1$ |
| c) $\log_3(x) = 3$ | g) $\log_2(x-1) = 0$ | k) $\log_3(2) = 0,25$ | o) $\log_2(2x - x^2) = 1$ |
| d) $\log_5(x) = -3$ | h) $\log(5x-1) = 1$ | l) $\log_3(81) = -4$ | p) $2 \log(x+5) = \log(x^2)$ |

Lösungen:**Aufgabe 1: Logarithmus**

- a) $\log_2(64) = 6$ c) $\log_4(256) = 4$ e) $\log_8(8) = 1$ g) $\log_{10}(0,001) = -3$ i) $\log_{36}(6) = \frac{1}{2}$
 b) $\log_3(27) = 3$ d) $\log_0(1) = 0$ f) $\log_3\left(\frac{1}{3}\right) = -1$ h) $\log_2\left(\frac{1}{32}\right) = -5$ j) $\log_{243}(3) = \frac{1}{5}$

Aufgabe 2: Logarithmus

- a) $3^4 = 81$ c) $6^2 = 36$ e) $8^0 = 1$ g) $3^2 = 9$ i) $9^{0,5} = 3$
 b) $4^3 = 64$ d) $2^6 = 64$ f) $5^1 = 5$ h) $7^2 = 49$ j) $64^{1/6} = 2$

Aufgabe 3: Logarithmus

- a) 4 c) 4 e) -3 g) -1 i) -3
 b) 3 d) -1 f) -1 h) -3 j) -8

Aufgabe 4: Logarithmengesetze

- a) $\log_2\left(\frac{a}{b}\right)$ d) $\frac{1}{8}$ g) 0
 b) $\log\left(1 + \frac{y}{x}\right)$ e) $\log(3)$ h) $\log\left(\frac{y}{x}\right)$
 c) $2 \cdot \log_2(b) - \log_2(c)$ f) $2 \log(x)$ i) 0

Aufgabe 5: Logarithmengesetze

- a) $\frac{\log(5)}{\log(3)} \approx 1,46$ c) $\frac{\log(100)}{\log(1,1)} \approx 48,3$ e) $3 \frac{\log(5)}{\log(2)} \approx 6,97$ g) $\log_2(x) - 3$ i) $\frac{1}{3}$
 b) $-\frac{\log(3)}{\log(2)} \approx 1,58$ d) $\frac{\log(24)}{\log(4)} \approx 2,29$ f) $5 - \frac{\log(3)}{\log(2)} \approx 3,41$ h) $-1,5 \log_2(s)$ j) $-1 - \log_2(y)$

Aufgabe 6: Logarithmengleichungen

- a) $D =]0; \infty[$, $L = \{9\}$ e) $D =]0; \infty[$, $L = \{1\}$ i) $D =]0; \infty[$, $L = \{3\}$ m) $D = \mathbb{R} \setminus \{-1; 1\}$, $L = \{-3; 3\}$
 b) $D =]0; \infty[$, $L = \{256\}$ f) $D =]0; \infty[$, $L = \{2\}$ j) $D =]0; \infty[$, $L = \{2\}$ n) $D = \mathbb{R}$, $L = \{\}$
 c) $D =]0; \infty[$, $L = \{27\}$ g) $D =]1; \infty[$, $L = \{2\}$ k) $D =]0; \infty[$, $L = \{16\}$ o) $D = [0; 2]$, $L = \{\}$
 d) $D =]0; \infty[$, $L = \{\frac{1}{125}\}$ h) $D =]\frac{1}{5}; \infty[$, $L = \{\frac{4}{5}\}$ l) $D =]0; \infty[$, $L = \{\frac{1}{3}\}$ p) $D =]-5; \infty[$, $L = \{-\frac{5}{2}\}$

Wie lauten die jeweiligen Fachbegriffe bzw. die Umkehrfunktion

$$\begin{array}{c}
 \text{Exponent} \\
 \boxed{2}^{\boxed{5}} = \boxed{32} \\
 \text{Term} \quad \text{Basis}
 \end{array}
 \begin{array}{c}
 \text{Ergebnis} \\
 \text{Potenzwert}
 \end{array}
 \Leftrightarrow \text{LOG}_{\boxed{2}}(\boxed{32}) = \boxed{5} \Leftrightarrow \frac{\text{LOG}(\boxed{32})}{\text{LOG}(\boxed{2})} = 5$$

Berechne den Logarithmus auf drei Nachkommastellen gerundet:

Beispiel: $2^{3x-5} + 6 = 134$

- Variable isolieren $2^{3x-5} = 128$
 - Logarithmieren $\lg(2^{3x-5}) = \lg 128$
 - Logarithmengesetze anwenden $(3x-5) \cdot \lg 2 = \lg 128 \quad | : \lg 2$
 - Nach Variable auflösen $3x-5 = \frac{\lg 128}{\lg 2} \quad | +5 | :3$
- $$x = \left(\frac{\lg 128}{\lg 2} + 5 \right) : 3 = 4$$

1) $7^{(x+2)} = 19$ - 0,48

2) $5 \cdot 2^{2x} = 10^{x+2}$ - 3,26

3) $\text{Log}_4(x \cdot 2) = 4$ 128

4) $8 \cdot 10^{x-1} = 5 \cdot 7^{3x}$

5) $5^x - 6^{x+1} = -6^x$

6) $\frac{1}{4^{x+1}} = \frac{2}{3}$

7) $4^{x-1} - 9^x = 3^{2x-1} - 2^{2x+1}$

$$1.) 7^{(x+2)} = 19$$

$$\Leftrightarrow \log_7(7^{x+2}) = \log_7(19)$$

$$\Leftrightarrow (x+2) \log_7(7) = \log_7(19)$$

$$\Leftrightarrow x+2 = \log_7(19)$$

$$\Leftrightarrow x = \log_7(19) - 2$$

$$\Leftrightarrow$$

$$2.) 5 \cdot 2^{2x} = 10^{x+2}$$

$$\Leftrightarrow 5 = \frac{10^{x+2}}{2^{2x}}$$

$$\Leftrightarrow \log_{10}(5 \cdot 2^{2x}) = \log_{10}(10^{x+2})$$

$$\Leftrightarrow \log_{10}(5) + \log_{10}(2^{2x}) = x+2$$

$$\Leftrightarrow \log_{10}(5) + 2x \cdot \log_{10}(2) = x+2$$

$$\Leftrightarrow \log_{10}(5) + 2x \log_{10}(2) = x+2$$

$$\Leftrightarrow \log_{10}(5) = x+2 - 2x \log_{10}(2)$$

$$\Leftrightarrow \log_{10}(5) - 2 = x(1 - 2 \log_{10}(2))$$

$$\Leftrightarrow x = \frac{\log_{10}(5) - 2}{1 - 2 \log_{10}(2)}$$

$$4.) 8 \cdot 10^{(x-1)} = 5 \cdot 7^{3x}$$

$$\Leftrightarrow \log_{10}(8 \cdot 10^{(x-1)}) = \log_{10}(5 \cdot 7^{3x})$$

$$\Leftrightarrow \log_{10}(8) + \log_{10}(10^{x-1}) = \log_{10}(5) + \log_{10}(7^{3x})$$

$$\Leftrightarrow \log_{10}(8) + (x-1) = \log_{10}(5) + (3x) \log_{10}(7)$$

$$\Leftrightarrow \log_{10}(8) - \log_{10}(5) - 1 = -x + 3x \log_{10}(7)$$

$$\Leftrightarrow \log_{10}\left(\frac{8}{5}\right) - 1 = x(-1 + 3 \log_{10}(7))$$

$$\Leftrightarrow \frac{\log_{10}\left(\frac{8}{5}\right) - 1}{3 \log_{10}(7) - 1} = x$$

$$\log_7(19) = y$$

$$\Leftrightarrow 7^y = 19 \quad \Leftrightarrow 7^{\log_7(19)} = 19$$

$$\Leftrightarrow \ln(7^y) = \ln(19) \quad \Leftrightarrow 19 = 7^y$$

$$\Leftrightarrow y \ln(7) = \ln(19)$$

$$\Leftrightarrow y = \frac{\ln(19)}{\ln(7)}$$

$$\log_7(19) = y$$

$$\Leftrightarrow 7^{\log_7(19)} = 19$$

$$\Leftrightarrow 19 = 7^y$$

$$\Leftrightarrow \ln(19) = \ln(7^y) = y \ln(7)$$

$$\Leftrightarrow y = \frac{\ln(19)}{\ln(7)}$$

$$3.) \log_4(x \cdot 2) = 4$$

$$\Leftrightarrow \log_4(x \cdot 2) = 4$$

$$\Leftrightarrow 2x = 4^4$$

$$\Leftrightarrow x = \frac{4^4}{2}$$

$$6^{x+1} = 6^x \cdot 6$$

$$5.) 5^x - 6^{x+1} = -6^x$$

$$\Leftrightarrow 5^x = -6^x + 6^{x+1} = -6^x + 6^x \cdot 6 = 6^x(-1+6)$$

$$\Leftrightarrow 5^x = 6^x \cdot 5$$

$$\Leftrightarrow \log_{10}(5^x) = \log_{10}(6^x \cdot 5)$$

$$\Leftrightarrow x \log_{10}(5) = \log_{10}(6^x) + \log_{10}(5)$$

$$\Leftrightarrow x \log_{10}(5) - \log_{10}(5) = x \log_{10}(6)$$

$$\Leftrightarrow \log_{10}(5)(x-1) = x \log_{10}(6)$$

$$\Leftrightarrow \frac{\log_{10}(5)}{\log_{10}(6)} = \frac{x}{x-1}$$

$$5.) 5^{-x} - 6^{x+1} = -6^x$$

~~$$\Leftrightarrow \log_{10}(5^{-x} - 6^{x+1}) = \log_{10}(-6^x)$$~~

$$\Leftrightarrow 5^{-x} = -6^x + 6^{x+1} = -6^x + 6^x \cdot 6 = 5 \cdot 6^x$$

$$\Leftrightarrow \log_5(5^{-x}) = \log_5(5 \cdot 6^x)$$

$$\Leftrightarrow x = \log_5(5) + \log_5(6^x)$$

$$\Leftrightarrow x = 1 + x \log_5(6)$$

$$\Leftrightarrow 1 = x - x \log_5(6)$$

$$\Leftrightarrow 1 = x(1 - \log_5(6))$$

$$\Leftrightarrow x = \frac{1}{1 - \log_5(6)}$$

$$\dots = \frac{1}{1 - \frac{\ln(6)}{\ln(5)}} = \frac{1}{\frac{\ln(5) - \ln(6)}{\ln(5)}} = \frac{\ln(5)}{\ln(5) - \ln(6)} = \frac{\ln(5)}{\ln\left(\frac{5}{6}\right)}$$

~~$$\log_a(b) = \log(b)$$~~

$$\log_a(b) = \frac{\log_2(b)}{\log_2(a)}$$

$$\log(a \cdot b) = \log(a) + \log(b)$$

$$6.) \frac{1}{4^{x-1}} = \frac{2}{3}$$

$$\Leftrightarrow \frac{3}{2} = 4^{x-1}$$

$$\Leftrightarrow \log_4\left(\frac{3}{2}\right) = \log_4(4^{x-1})$$

$$\Leftrightarrow \log_4\left(\frac{3}{2}\right) = (x-1)$$

$$\Leftrightarrow x = \log_4\left(\frac{3}{2}\right) + 1$$

$$= \log_4(3) - \log_4(2) + 1$$

$$= \frac{\log(3)}{\log(4)} - \frac{\log(2)}{\log(4)} + 1$$

$$= \frac{\log(3) - \log(2)}{\log(4)} + 1$$

$$7.) 4^{x-1} - 9^x = 3^{2x-1} - 2^{2x+1}$$

$$\Leftrightarrow (2^2)^{x-1} - (3^2)^x = 3^{2x-1} - 2^{2x+1}$$

$$\Leftrightarrow 2^{2x+2} - 3^{2x} = 3^{2x-1} - 2^{2x+1}$$

$$\Leftrightarrow 2^{2x+2} + 2^{2x+1} = 3^{2x-1} + 3^{2x}$$

$$\Leftrightarrow 2^{2x+1} \cdot 2 + 2^{2x+1} = 3^{2x} \cdot \frac{1}{3} + 3^{2x}$$

$$\Leftrightarrow 2^{2x+1}(2+1) = 3^{2x}\left(\frac{1}{3} + 1\right)$$

$$\Leftrightarrow 2 \cdot 2^{2x}(3) = 3^{2x} \cdot \frac{4}{3}$$

$$\Leftrightarrow 6 \cdot 2^{2x} = \frac{4}{3} 3^{2x}$$

$$\Leftrightarrow \frac{9}{2} \cdot 2^{2x} = 3^{2x}$$

$$\Leftrightarrow \frac{9}{2} = \frac{3^{2x}}{2^{2x}}$$

$$\Leftrightarrow \log\left(\frac{9}{2}\right) = \log\left(\frac{3^{2x}}{2^{2x}}\right)$$

$$\text{z.z. } x := \log_a(b) = \frac{\log_n(a)}{\log_n(b)}$$

$$\log_a(b) = x$$

$$\Leftrightarrow x = \log_a(b)$$

$$\Leftrightarrow a^x = a^{\log_a(b)} = b$$

$$\log_a(b) = x$$

$$\Leftrightarrow a^x - b = 0$$

$$\Leftrightarrow a^x - b + x = x$$

$$\Leftrightarrow \log_n(a^x) = \log_n(b)$$

$$\Leftrightarrow x \log_n(a) = \log_n(b)$$

$$\Leftrightarrow x = \frac{\log_n(b)}{\log_n(a)}$$

log a

$$\log_a(b) = x$$

$$\Leftrightarrow a^{\log_a(b)} = a^x$$

$$\Leftrightarrow b = a^x$$

$$\Leftrightarrow \ln(b) = \ln(a^x)$$

$$\Leftrightarrow \ln(b) = x \ln(a)$$

$$\Leftrightarrow x = \frac{\ln(b)}{\ln(a)} = \frac{\log_a(b)}{\log_a(a)} = \frac{\log_a(b)}{1} = \log_a(b)$$

$$= \frac{\log_b(b)}{\log_b(a)} = \frac{1}{\log_b(a)}$$

log b

$$\ell_n(x) := \log_e(x)$$

$$x = \log_a(b)$$

$$\Leftrightarrow a^x = b$$

$$\Leftrightarrow \log_n(a^x) = \log_n(b)$$

$$\Leftrightarrow x \log_n(a) = \log_n(b)$$

$$\Leftrightarrow x = \frac{\log_n(b)}{\log_n(a)}$$

$$\log_4(11) = \frac{\log_n(11)}{\log_n(4)} = \frac{\ln(11)}{\ln(4)}$$

$$\log_3(30) = x$$

$$\Leftrightarrow \log_n(x)$$

$$\log_a(a) = 1 \frac{2}{2}$$

$$\log_a(b^c) = d$$

$$\Leftrightarrow a^{\log_a(b^c)} = a^d$$

$$\Leftrightarrow b^c = a^d$$

$$\Leftrightarrow \log_b(b^c) = \log_b(a^d)$$

$$\Leftrightarrow c \cdot \log_b(b) = \log_b(a^d)$$

$$\Leftrightarrow c = \log_b(a^d)$$

$$\text{Fall: } c = d = 1$$

$$\log_a(b^1) = 1 \Leftrightarrow 1 = \log_a(a)$$

$$\Leftrightarrow \log_a(b) = \log_b(a) = 1$$

$$a.3.1$$

$$e.) 2^x = 7 \quad x = ?$$

$$\Leftrightarrow \log_2(2^x) = \log_2(7)$$

$$\Leftrightarrow x = \log_2(7)$$

$$f.) 7^{x+1} = 11$$

$$\Leftrightarrow \log_7(7^{x+1}) = \log_7(11)$$

$$\Leftrightarrow \log_7(7^{x+1}) = \log_7(11)$$

$$\Leftrightarrow (x+1) \cdot \log_7(7) = \log_7(11)$$

$$\Leftrightarrow x+1 = \log_7(11)$$

$$\Leftrightarrow x = \log_7(11) - 1$$

$$\log_a(b^c) = c \cdot \log_a(b)$$

$$g.) 3^x = 12$$

$$\Leftrightarrow \log_3(3^x) = \log_3(12)$$

$$\Leftrightarrow x \cdot \log_3(3) = \log_3(12)$$

$$\Leftrightarrow x = \log_3(12)$$

$$\approx 2,26$$

$$h.) 5^{x-2} = 3$$

$$\Leftrightarrow \log_5(5^{x-2}) = \log_5(3)$$

$$\Leftrightarrow (x-2) \underbrace{(\log_5(5))}_{=1} = \log_5(3)$$

$$\Leftrightarrow x-2 = \log_5(3)$$

$$\Leftrightarrow x = \log_5(3) + 2$$

$$\approx 2,68$$

$$i.) 2^{-x} = 20$$

$$\Leftrightarrow \log_2(2^{-x}) = \log_2(20)$$

$$\Leftrightarrow -x \log_2(2) = \log_2(20)$$

$$\Leftrightarrow +x = -\log_2(20)$$

$$\approx -4,32$$

$$k.) \frac{1}{4^{x+1}} = \frac{2}{3} \quad | \cdot 3 : 2 \cdot 4^{x+1}$$

$$\frac{3}{2} = 4^{x+1}$$

$$\Leftrightarrow \log_4(4^{x+1}) = \log_4\left(\frac{3}{2}\right)$$

$$\Leftrightarrow (x+1) \log_4(4) = \log_4\left(\frac{3}{2}\right)$$

$$\Leftrightarrow x = \log_4\left(\frac{3}{2}\right) - 1$$

$$\approx -0,71$$

$$x = \log_a(b) = \frac{\log_n(b)}{\log_n(a)}$$

AMA

$$\log_a(b) = c$$

$$\Leftrightarrow a^{\log_a(b)} = a^c$$

$$\Leftrightarrow b = a^c$$

$$1) 7^{x+2} = 19$$

$$\Leftrightarrow \log_7(7^{x+2}) = \log_7(19)$$

$$\Leftrightarrow (x+2) \log_7(7) = \log_7(19)$$

$$\Leftrightarrow x+2 = \log_7(19)$$

$$\Leftrightarrow x = \log_7(19) - 2$$

$$2) 5 \cdot 2^{2x} = 10^{x+2}$$

$$\Leftrightarrow \log_2(5 \cdot 2^{2x}) = \log_2(10^{x+2})$$

$$\Leftrightarrow \log_2(5) + \log_2(2^{2x}) = (x+2) \log_2(10)$$

$$\Leftrightarrow \log_2(5) + 2x \log_2(2) = (x+2) \log_2(10)$$

$$\Leftrightarrow x = 2x \log_2(10) + 2 \log_2(10) - \log_2(5)$$

$$\Leftrightarrow x - 2x \log_2(10) = 2 \log_2(10) - \log_2(5)$$

$$\Leftrightarrow x(1 - 2 \log_2(10)) = 2 \log_2(10) - \log_2(5)$$

$$\Leftrightarrow x = \frac{2 \log_2(10) - \log_2(5)}{1 - 2 \log_2(10)} \approx -1,86$$

$$\log_2 \approx -3,26$$

$$3) \log_4(x \cdot 2) = 4$$

$$\Leftrightarrow 4^{\log_4(x \cdot 2)} = 4^4$$

$$\Leftrightarrow x \cdot 2 = 256$$

$$\Leftrightarrow x = 128$$

$$4) 8 \cdot 10^{x-1} = 5 \cdot 7^{3x}$$

$$\Leftrightarrow \log_{10}(8 \cdot 10^{x-1}) = \log_{10}(5 \cdot 7^{3x})$$

$$\Leftrightarrow \log_{10}(8) + \log_{10}(10^{x-1}) = \log_{10}(5) + \log_{10}(7^{3x})$$

$$\Leftrightarrow \log_{10}(8) + (x-1) = \log_{10}(5) + 3x \log_{10}(7)$$

$$\Leftrightarrow x-1 - 3x \log_{10}(7) = \log_{10}(5) - \log_{10}(8)$$

$$\Leftrightarrow x(1 - 3 \log_{10}(7)) = 1 + \log_{10}\left(\frac{5}{8}\right)$$

$$\Leftrightarrow x = \frac{1 + \log_{10}\left(\frac{5}{8}\right)}{1 - 3 \log_{10}(7)} \approx -0,518$$

$$z.z. \log_a(x^n) = n \cdot \log_a(x)$$

$$y := \log_a(x^n)$$

$$\log_a(x^n) =: y$$

$$\Leftrightarrow a^y = a^{\log_a(x^n)} = x^n$$

$$\log_a(x^n) = \dots$$

$$\Leftrightarrow a^y = x^n$$

letz

$$\boxed{\log_a(x^n) = n \cdot \log_a(x)}$$

$$\text{sei } x \stackrel{!(1)}{=} a^m \Leftrightarrow m \stackrel{!(2)}{=} \log_a(x)$$

$$y := \log_a(x^n)$$

$$y = \log_a(x^n) \stackrel{!(1)}{=} \log_a((a^m)^n) = \log_a(a^{m \cdot n}) = m \cdot n \stackrel{!(2)}{=} \log_a(x) \cdot n = n \cdot \log_a(x)$$

$$a) \log_2(\sqrt{5}) = \log_2(5^{\frac{1}{2}}) = \frac{1}{2} \log_2(5)$$

$$b) \log_a(\sqrt[3]{a^4}) = \log_a(a^{\frac{4}{3}}) = \frac{4}{3} \cdot \log_a(a) = \frac{4}{3}$$

$$c) \log_a\left(\frac{1}{x}\right) = \log_a(x^{-1}) = -1 \log_a(x) = -\log_a(x)$$

$$d) \log_2\left(\frac{1}{\sqrt[4]{48}}\right) = \log_2(48^{-\frac{1}{4}}) = -\frac{1}{4} \log_2(48)$$

$$e) \log_5\left(\frac{81}{\sqrt[3]{3}}\right) = \log_5\left(\frac{3^4}{3^{\frac{1}{3}}}\right) = \log_5\left(3^{4 - \frac{1}{3}}\right) = \log_5\left(3^{\frac{12-1}{3}}\right) = \log_5\left(3^{\frac{11}{3}}\right) = \frac{11}{3} \log_5(3)$$

$$f) \log_2(\sqrt{8} \cdot \sqrt[3]{16}) = \log_2(\sqrt{2^3} \cdot \sqrt[3]{2^4}) = \log_2(2^{\frac{3}{2}} \cdot 2^{\frac{4}{3}}) = \log_2(2^{\frac{3}{2} + \frac{4}{3}}) = \log_2(2^{\frac{17}{6}}) = \frac{17}{6} \log_2(2) = \frac{17}{6} \cdot 1 = \frac{17}{6}$$

q.e.d.

$$81 = 9 \cdot 9 = 3^2 \cdot 3^2 = 3^{2+2} = 3^4$$

$$8 = 2 \cdot 2 \cdot 2 = 2^3$$

$$16 = 2^4$$

7.7: ~~log~~

$$(a^m)^n = a^{m \cdot n}$$

sei $a^n =: x$

$$\Rightarrow (a^n)^m = x^m = \underbrace{x \cdot x \cdot x \cdot \dots \cdot x}_{m\text{-mal}} \quad \text{A}$$

$$\stackrel{\text{A}}{=} a^n \cdot a^n \cdot a^n \cdot \dots \cdot a^n$$

$$= \underbrace{(\underbrace{a \cdot \dots \cdot a}_{n\text{-mal}}) \cdot (\underbrace{a \cdot \dots \cdot a}_{n\text{-mal}}) \cdot (\dots) \cdot (\underbrace{a \cdot \dots \cdot a}_{n\text{-mal}})}_{m\text{-mal}}$$

$$x = a^n = \underbrace{a \cdot a \cdot a \cdot \dots \cdot a}_{n\text{-mal}}$$

$$\Rightarrow \text{man hat } n\text{-mal}(\text{A } n\text{-mal}(a))$$

$$y := a^n$$

$$= \underbrace{(\underbrace{a \cdot \dots \cdot a}_{n\text{-mal}}) \cdot (\underbrace{a \cdot \dots \cdot a}_{n\text{-mal}}) \cdot (\dots) \cdot (\underbrace{a \cdot \dots \cdot a}_{n\text{-mal}})}_{m\text{-mal}}$$

$$= \underbrace{y \cdot y \cdot \dots \cdot y}_{m\text{-mal}} = y^m = (a^n)^m$$

$$\text{d.h.: } (a^m)^n = (a^n)^m = a^{m \cdot n} = a^{n \cdot m}$$

$$\log_a((a^m)^n) = \log_a(a^{m \cdot n}) = m \cdot n$$